Economic Representations

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Overview

1. Introduction

2. Design

3. Result

4. Ideas

5. Appendix

Summary

(a) Language Models

Train → LLM

- Train language models on historical news data
- Time-stamped training data prevents lookahead bias from pretraining

(b) Representations

WSJ, B BARRON'S, MW

LLM → $x_{s,t}$

- Compute an embedding vector for each firm in each year
- Contrastive learning targets vector representations to language about firms

(c) Outcome Functions

$x_{s,t} \xrightarrow{f_t} Y_{s,t}$

- Estimate functions that map representations to outcomes
- Sample splitting strategy evaluates estimates on unseen data

Question

- Q1: How can we measure the market's perception of each firm?
- Q2: How do these perceptions impact firm valuation and stock returns?

Motivation

- **Market's perception** is crucial for **firm value** (Mullainathan et al., 2008, QJE)
  - The market's perception of a firm's **business model**
  - E.g., Tesla: "internet-of-cars company" vs. "just an automaker"
  - Fundamental impact on the market's expectations
- **Challenge**: This **subjective belief** is hard to measure using **structured data**
- **News** text is informative about **market's perception** (Zhou et al., 2024, MS)
  - Journalists and investors describe firms in common ways
  - The press has incentives to match how its readers think
- **This paper uses a LLM to quantify market's perception from news**

Marginal Contribution

- The impact of market's perception on firm value
  - Prior studies
    - **Theoretically** demonstrate the importance of market perception for firm value (Mullainathan et al., 2008, QJE; Schwartzstein and Sunderam, 2021, AER; Bohren et al., 2023, NBER)
  - This paper
    - **Employ an LLM algorithm to quantify** market perception and empirically demonstrate its impact on firm value

Hypothesis

- H1: **Higher representation similarity** implies **higher return correlation and shared analyst probability**
- H2: Representations improve the ability to **explain firms' valuations and stock returns**
- H3: The **abnorm deviations in representation** reflect **mispricing**
  - Mean reversion in representation $\Longrightarrow$ Correction in stock price

Measuring Representations from Language

- Step 1: Pretrain a **general-purpose** language model (RepresentLM)
  - BERT-based architecture
  - Use American news articles from 1900 to 1979 (not restricted to financial news)
  - Convert articles into representations vectors ($K = 768$)
- **Drawback**: Fail to capture **firm heterogeneity**

Measuring Representations from Language

- Step 2: Fine-tune the RepresentLM using contrastive training objective:

$$\min_{\theta \in \Theta} -\frac{1}{n} \sum_{i=1}^{n} \log \frac{\exp\{\psi(e(a_i;\theta), e(b_i;\theta))/\tau\}}{\sum_{j=1}^{n} \exp\{\psi(e(a_i;\theta), e(b_j;\theta))/\tau\}} \tag{1}$$

- a_i and b_i are two different financial news for a same frim i
- $e(\cdot;\theta_t)$ and $\psi(\cdot,\cdot)$ represent embedding vector and **cosine similarity**, respectively

- **Generate vectors for the same firm more similar** than those for different firms
- **Fine-tune in each year** from 1979 to 2022 to **avoid lookahead bias**

Measuring Representations from Language

- Step 3: Computing firm-level representations
  - Use financial news articles
  - Compute embedding vectors for year t articles using the model fine-tuned in year $t-1$
  - Average article vectors monthly, then average monthly embeddings annually
  - **Firm-year-level representation vectors**

- **These vectors quantify the market's perception of the firm's business model**

Split-Sample Procedure for Empirical Tests

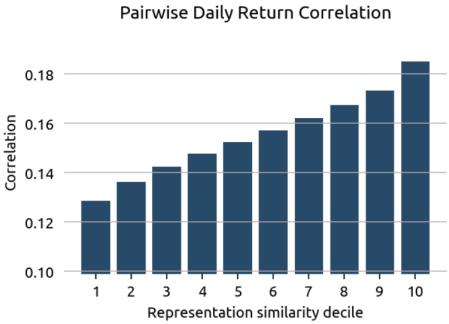
- Use **Ridge to estimate fitted outcome variables** $\widehat{Y}_{s,t}$ based on representation vectors $x_{s,t}$, then **use** $\widehat{Y}_{s,t}$ **to explain** $Y_{s,t}$ **(**R^2**)**
- Estimate hyperparameters of Ridge using five-fold cross validation
  - Construct the five folds by randomly sampling folds at the firm (permco) level
  - Tune hyperparameters on the **other four folds within each year** t
- Prevent using the firm's own data to estimate hyperparameters, thereby **mitigating overfitting and inflated** R^2

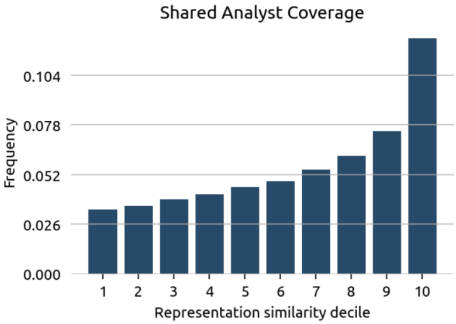
Data

- News: Dow Jones Newswires
- Stock return: CRSP
- Firm characteristics: Data set of Jensen et al. (2023)
- Time period for empirical tests: 1984–2021

Measure Validation

- H1: **Higher representation similarity** implies **higher return correlation and shared analyst probability**





Helping to Explain Firms' Valuations

- H2a: Representations improve the ability to **explain firms' valuations**

R^2: Estimates of price and forecast variables			
Dependent variable →	log(P/B)	log(P/E)	$\mathbb{F}$[LTG]
Explanatory variables ↓	(1)	(2)	(3)
Representation	26.0%	12.1%	19.7%
Industry (FF12)	9.2%	6.8%	12.8%
Sub-Industry (FF48)	11.7%	7.5%	13.6%
Characteristics	12.8%	6.9%	22.2%
Representation+All Others	32.9%	14.5%	30.1%
All Others	19.3%	10.7%	25.7%

Helping to Explain Firms' Stock Returns: Theoretical Hypothesis

- **Representations and prices**: The stock price of firm s at time t is

$$\ln(P_{s,t}) = v_t \cdot x_{s,t} + \eta_{s,t}, \tag{2}$$

- $x_{s,t}$ is the market's perception of the firm
- v_t is how the market prices the firm's representation (valuation function)
- $\eta_{s,t}$ is other features that cannot be explained by $x_{s,t}$ and v_t

Helping to Explain Firms' Stock Returns: Theoretical Hypothesis

- **Stock Return**: The change in stock price of firm s from time t to $t+1$ is

$$R_{s,t+1} \approx \underbrace{\Delta v_{t+1} \cdot x_{s,t}}_{\text{valuation function change}} + \underbrace{v_{t+1} \cdot \Delta x_{s,t+1}}_{\text{representation change}} + \varepsilon_{s,t+1} \tag{3}$$

- where $\Delta v_{t+1} \equiv v_{t+1} - v_t$ and $\Delta x_{s,t+1} \equiv x_{s,t+1} - x_{s,t}$ are changes in the valuation function and representation

Helping to Explain Firms' Stock Returns: Empirical Tests

- H2b: Representations improve the ability to **explain firms' realized stock returns**

R^2: Estimates of annual returns			
Estimation strategy →	[Δv]	[Δx]	[Total]
Explanatory variables ↓	(1)	(2)	(3)
Representation	9.4%	6.8%	13.4%
Industry (FF12)	3.5%	<0%	3.5%
Sub-Industry (FF48)	4.1%	<0%	4.1%
Characteristics	3.1%	4.4%	8.6%
Representation+All Others	11.5%	10.0%	18.7%
All Others	6.4%	4.4%	11.3%

Reversion in Representation and Mispricing

- **Define the abnormal deviation in representation**

$$\Delta x^h_{s,t} \equiv x_{s,t} - \overline{x}^h_{s,t} \tag{4}$$

- $\overline{x}^h_{s,t}$ is the average of the firm's representation over the previous h years (h=5)

- **Deviations in representation revert**

$$\text{Reversion}_{s,t+1} = -\psi(\Delta x_{s,t+1}, \Delta x^h_{s,t})$$

- The author finds that mean reversion is **0.38** $\Longrightarrow$ **Mean reversion** in representation

Reversion in Representation and Mispricing

- Fama–MacBeth regression: $R_{s,m,t+1} = \alpha_m + \beta_m^T Z_{s,t} + \varepsilon_{s,m,t+1}$
- Representation Deviation (priced): $\widehat{v}_t \cdot \Delta x_{s,t}^h$
- H3: The **abnorm deviations in representation** reflect **mispricing**

Dependent variable: Monthly return [%]					
	(1)	(2)	(3)	(4)	(5)
Representation Deviation (priced)	-0.45***	-0.46***	-0.28***	-0.30**	-0.27***
	(0.16)	(0.14)	(0.10)	(0.13)	(0.10)
Valuation Function Deviation (priced)	0.06	0.18	0.24	0.18	0.26
	(0.24)	(0.19)	(0.19)	(0.21)	(0.16)
Deviation in $\log(P/B)$	0.08				-0.14
	(0.24)				(0.19)
$\text{Return}_{12,1}$		0.04			0.37
		(0.31)			(0.24)
$\text{Return}_{36,1}$			-0.52		-0.46
			(0.40)		(0.36)
$\log(P/B)$				-0.52	-0.35
				(0.32)	(0.27)
Forecasting R^2	0.013	0.016	0.017	0.015	0.028
Months	456	456	456	456	456

Extension

1. Leverage LDA to investigate which thematic content contributes to misrepresentation
   - The impact of textual topics on analysts' cognitive biases (Ke, 2026, AFA; Bastianello et al., 2026, AFA)

2. Quantify the representations held by different investor classes
   - Analysts: Analyst reports
   - Retail Investors: Reddit, Twitter, and StockTwits

Estimating Components of Returns

- **Valuation function change component**

$$R_{s',t+1} = \delta_{t+1} \cdot x_{s',t} + \varepsilon_{s',t+1} \tag{5}$$

$$\widehat{[\Delta v]}_{s,t+1} = \widehat{\delta}_{t+1} \cdot x_{s,t} \tag{6}$$

- **Representation change component**

$$R_{s',t+1} = \nu_{t+1} \cdot \Delta x_{s',t+1} + \varepsilon_{s',t+1} \tag{7}$$

$$\widehat{[\Delta x]}_{s,t+1} = \widehat{\nu}_{t+1} \cdot \Delta x_{s,t+1} \tag{8}$$

Estimating Components of Returns

- **Total estimated return**

$$R_{s',t+1} = \zeta_{t+1} \cdot c_{s',t+1} + \varepsilon_{s',t+1} \tag{9}$$

$$\widehat{[\text{Total}]}_{s,t+1} = \widehat{\zeta}_{t+1} \cdot c_{s,t+1} \tag{10}$$

- $c_{s,t+1} \equiv [x_{s,t} \quad \Delta x_{s,t+1}]$

10-K Filing Language and Returns

- Market participants may not always pay attention to changes in 10-K filings
- News language has additional information that could be useful for measuring the market's perceived business model

R^2: Estimates of annual returns		
Estimation strategy →	$[\Delta v]$	$[\Delta x]$
Explanatory variables ↓	(1)	(2)
Representation	9.3%	7.1%
10-K Filing	3.3%	0.4%

Manager Communication and Misrepresentation

- Communication by managers is associated with the market's perceptions

	Representation Deviation (priced)
	(1)
Managerial Communication Deviation (priced)	0.146***
	(0.048)
Intercept	0.427***
	(0.024)
Years	18
R^2	0.061